

PEM-CAN: Parameter-Efficient Multimodal Fusion for Diabetic Neuropathy Screening: Architectural Framework and Simulation Benchmarking

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Abstract—Screening for diabetic autonomic neuropathy (DAN) requires bridging spatial ocular pathology with continuous ambulatory physiological signals. However, pervasive digital health gateways face strict on-device memory budgets, telemetry dropout, and institutional data governance boundaries. Here, we present the Parameter-Efficient Multimodal Cross-Attention Network (PEM-CAN), an architectural framework designed to fuse spatial retinal representations with high-frequency continuous glucose monitoring (CGM) and wearable actigraphy telemetry. Evaluated on a statistically calibrated simulation benchmark modeling the NIH Bridge2AI AI-READI Type 2 Diabetes protocol ($N = 2,840$ patient profiles), PEM-CAN freezes foundation backbones and restricts adaptation to 2.60 M parameters (1.75% of the 148.6 M model) via low-rank cross-attention constrained by Stiefel-manifold orthogonality penalties. On the held-out test split ($N = 568$, 128 positives, 22.5% DAN prevalence), PEM-CAN achieves an AUROC of 0.934 [95% CI: 0.904, 0.960], an AUPRC of 0.795 ± 0.011 , an F1-score of 0.803 ± 0.012 , and an Expected Calibration Error of 0.038 at 90.5% accuracy—strictly bounded below the theoretical Bayes optimal oracle ceiling of 0.957 AUROC. Progressive telemetry dropout confirms resilience (0.861 AUROC under 50% packet loss). Hardware profiling demonstrates 20.2 ms inference latency on NVIDIA Jetson Orin Nano, while federated optimization via FFA-LoRA transmits 10.4 MB/round (98.25% bandwidth reduction). Our benchmark establishes an open, reproducible framework for parameter-efficient multimodal sensing.

Index Terms—Parameter-Efficient Fine-Tuning (PEFT), Multimodal Deep Learning, Diabetic Autonomic Neuropathy, Retinal Fundus Imaging (Oculomics), Continuous Glucose Monitoring (CGM), Wearable Biosensors, Stiefel Manifold Regularization, Federated Health, Edge Artificial Intelligence.

Translational Relevance—This study presents a parameter-efficient foundation fusion framework and simulation benchmark for non-invasive diabetic autonomic neuropathy screening in ambulatory healthcare. By constraining cross-modal attention between spatial retinal photographs and continuous wearable sensor streams to a Stiefel-manifold orthogonal low-rank subspace, PEM-CAN achieves strong diagnostic discrimination (0.934 AUROC) while training only 1.75% of model weights. The architecture maintains stability under 50% wearable packet loss, preserves linear computational memory scaling, and slashes federated communication overhead by over 98%, establishing a reproducible benchmark for multi-organ clinical AI deployment.

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I. INTRODUCTION

DIABETIC autonomic neuropathy (DAN) is among the most clinically devastating yet frequently overlooked microvascular complications of diabetes mellitus, doubling the five-year mortality rate of affected patients through silent myocardial infarction, malignant cardiac arrhythmias, and sudden metabolic decompensation [1]–[3]. Despite its lethal trajectory, clinical diagnosis remains largely reactive [4]. Standard staging depends on specialized cardiovascular autonomic reflex tests (CARTs) or sudomotor axon reflex evaluations, which require dedicated autonomic laboratory infrastructure and are rarely performed during routine outpatient visits [5]. Furthermore, conventional glycemic monitoring via glycated hemoglobin (HbA1c) represents a blunt retrospective instrument; by condensing three months of complex metabolic fluctuations into a single blurred mean, HbA1c remains blind to rapid postprandial glucose surges, nocturnal hypoglycemic crashes, and circadian autonomic variability that directly accelerate microvascular and neurovascular damage [12], [13].

To transcend these diagnostic blind spots, biomedical informatics must unify two distinct observational windows into human physiology [14], [16]:

- 1) **The Ocular Window (Cumulative Structural Damage):** The retina is embryologically and anatomically an exposed extension of the central nervous system [9]. High-resolution retinal fundus photography non-invasively captures the microvascular architecture of the human body, revealing capillary rarefaction, arteriolar narrowing, and microaneurysms that correlate directly with systemic autonomic and peripheral nerve fiber attrition [6]–[8], [10], [11].
- 2) **The Wearable Window (Dynamic Physiological Provocation):** Continuous glucose monitoring (CGM) and commercial wearable actigraphy capture the dynamic metabolic and autonomic excursions that drive microvascular damage [12], [14]. Wearable sensor telemetry provides longitudinal recordings of interstitial glucose fluctuations, resting heart rate volatility, respiratory patterns, sleep architecture, and autonomic stress scores over days and weeks.

The NIH Bridge2AI AI-READI initiative establishes an equitable multi-organ data standard coupling macula- and disc-centered fundus photographs with continuous Dexcom G6 glucose telemetry and Garmin Vivosmart 5 wristband actigraphy across diverse Type 2 Diabetes cohorts [15]. While

access to raw clinical repositories is governed by institutional review boards and data use agreements, open and statistically calibrated simulation benchmarks provide a mathematically rigorous testbed to evaluate multimodal fusion architectures under controlled physiological signal-to-noise ratios.

Translating coupled ocular-wearable telemetry into deployable clinical artificial intelligence exposes four key engineering hurdles [16], [18]:

- **Modality Asymmetry and Gradient Dominance:** Retinal photography provides static, high-entropy 2D spatial pixel arrays ($384 \times 384 \times 3$), whereas wearable telemetry produces non-stationary 1D temporal sequences. When standard deep neural networks fuse these inputs, large visual gradient magnitudes overpower subtle physiological trends—a failure mode known as modality dominance [17], [18]. Conversely, naive late decision fusion discards fine-grained cross-modal token correlations entirely.
- **Training Memory Scaling across Temporal Horizons:** Concatenating temporal and visual tokens in dense self-attention architectures induces quadratic attention memory expansion $\mathcal{O}((M + L)^2)$ [19]. As surveillance horizons scale from days to weeks ($L > 2,000$ tokens), full fine-tuning of massive vision backbones triggers severe out-of-memory faults on point-of-care workstations.
- **Pervasive Sensor Telemetry Missingness:** In real-world ambulatory environments, wearable sensors routinely suffer 10% to 50% data loss due to battery exhaustion, sleep compression, or sensor detachment [18]. Standard multimodal models degrade precipitously when time-series tokens drop out.
- **Decentralized Privacy and Bandwidth Constraints:** Hospital privacy regulations prohibit aggregating raw retinal images and wearable telemetry into central repositories, requiring federated optimization where full-model parameter exchange (> 500 MB/round) exceeds bandwidth limits [30]–[32].

To overcome these roadblocks, we present the **Parameter-Efficient Multimodal Cross-Attention Network (PEM-CAN)**. PEM-CAN freezes pre-trained foundation visual backbones, injecting lightweight low-rank adapters ($r = 8$) directly into bidirectional cross-attention projections under a Stiefel-manifold orthogonal regularizer [24], [29]. By constraining cross-modal interaction to an orthogonal low-rank subspace, PEM-CAN mitigates modality collapse, operates with linear memory complexity $\mathcal{O}(M \cdot L)$, and maintains high diagnostic fidelity under severe sensor missingness.

The core contributions of this study are fourfold:

- 1) **Ocular-Wearable Fusion Framework:** We formulate a parameter-efficient foundation fusion architecture that synchronizes spatial retinal fundus photography with multi-day continuous glucose monitoring and wearable actigraphy, handling dimensional and sampling asymmetries without fine-tuning heavy visual backbones.
- 2) **Stiefel-Manifold Orthogonal Regularization:** We introduce a low-rank adapter projection constrained by a Stiefel-manifold orthogonality penalty ($\mathcal{L}_{\text{orth}}$), ensuring that low-rank basis vectors remain well-conditioned and

preventing visual feature dominance.

- 3) **Calibrated Benchmark on AI-READI Protocol:** We establish a 5-fold cross-validated benchmark on $N = 2,840$ patient records statistically calibrated to the NIH Bridge2AI AI-READI protocol across 19 competing clinical, heuristic, and foundation baselines. PEM-CAN achieves superior diagnostic discrimination (0.934 AU-ROC, 0.803 F1-score, 0.038 ECE at 22.5% prevalence)—strictly bounded below the theoretical Bayes optimal oracle ceiling of 0.957 AUROC.
- 4) **Edge Profiling and Federated Scaling:** We validate edge execution (20.2 ms latency, 1.18 GB FP16 footprint on NVIDIA Jetson Orin Nano) and demonstrate up to 99.56% reduction in federated client transmission bandwidth across 50 decentralized hospital nodes.

II. RELATED WORK

A. Multimodal Deep Learning in Diabetes and Oculomics

Deep learning has achieved notable success in ophthalmic disease screening, from diabetic retinopathy classification [7], [8] to systemic biomarker prediction from retinal photography [6], [9]. The release of retinal foundation models such as RETFound [10] demonstrated that self-supervised pre-training on large fundus repositories yields transferable representations. Concurrently, continuous glucose monitoring and wearable biosensors have been used to quantify glycemic variability (Time in Range, Mean Amplitude of Glycemic Excursions) and predict metabolic parameters [12]–[14]. However, prior work largely analyzes retinal imaging and physiological biosignals in isolation or through late decision fusion. Joint end-to-end architectures that explicitly correlate spatial microvascular lesions with continuous temporal metabolic stress remain largely unexplored.

B. Parameter-Efficient Fine-Tuning (PEFT)

To adapt massive foundation models without full retraining, parameter-efficient fine-tuning (PEFT) methods update small auxiliary modules while freezing backbone weights. Prominent approaches include Low-Rank Adaptation (LoRA) [24], which decomposes weight updates into low-rank factor matrices $\Delta W = BA$, Weight-Decomposed LoRA (DoRA), Bottleneck Adapters [25], BitFit (bias-only tuning) [26], and dynamic-rank allocation (AdaLoRA) [27]. In multimodal applications, adapters have primarily been explored for vision-language tasks. PEM-CAN extends low-rank adaptation to bidirectional cross-modal attention between spatial 2D images and 1D longitudinal biosignals, introducing Stiefel-manifold orthogonal constraints [29] to prevent visual modality dominance.

C. Missing-Modality Learning and Gated Fusion

Ambulatory wearable biosensors regularly suffer from sensor detachment, motion artifacts, and transmission dropouts [18]. Architectural strategies for handling missing modalities include Gated Multimodal Units (GMU) [22], Multimodal Transfer Modules (MMTM) [23], and cross-modal attention transformers

such as MuT [20] and Perceiver IO [21]. However, full fine-tuning of these models incurs substantial memory overhead during multi-day surveillance training. PEM-CAN provides an alternative by pairing frozen unimodal representations with low-rank cross-attention adapters that maintain graceful performance degradation under up to 50% sensor packet loss.

III. PROBLEM FORMULATION AND SYSTEM ARCHITECTURE

A. Mathematical Problem Formulation

Let $(\Omega, \mathcal{F}, \mathbb{P})$ denote a probability space representing patient physiology. A patient produces an observation tuple $X = (X_v, X_g, X_a) \in \mathcal{X}_v \times \mathcal{X}_g \times \mathcal{X}_a$. The visual space $\mathcal{X}_v \subseteq \mathbb{R}^{H \times W \times 3}$ comprises macula-centered retinal fundus photography ($H = W = 384$). The metabolic time-series space $\mathcal{X}_g \subseteq \mathbb{R}^{T_g \times 1}$ encapsulates interstitial glucose values recorded via continuous glucose monitoring at 5-minute sampling intervals ($T_g = 2,016$ steps across 7 days, scaling up to $T_g = 8,640$ steps across 30 days) [12], [13]. The autonomic actigraphy space $\mathcal{X}_a \subseteq \mathbb{R}^{T_a \times D_a}$ encapsulates wearable sensor telemetry mapped to the Garmin Vivosmart 5 schema ($D_a = 5$ channels: resting heart rate, respiration rate, device stress score, step counts, and sleep metrics) [15].

The diagnostic objective is to estimate the posterior probability for a multi-task clinical target vector $Y = [y_{\text{DAN}}, y_{\text{DR}}, y_{\text{CKD}}, y_{\text{DPN}}]^\top \in \{0, 1\}^4$:

$$P(Y | X_v, X_g, X_a) = \int_{\mathcal{Z}} P(Y | Z) P(Z | X_v, X_g, X_a) dZ \quad (1)$$

where $\mathcal{Z} \subseteq \mathbb{R}^{d_m}$ is a shared latent representation manifold ($d_m = 768$). Target leakage is strictly prevented: no prediction endpoint is computed directly from input time-series metrics.

To ensure that multimodal fusion yields non-trivial diagnostic utility beyond unimodal representations, the learning objective is formulated to capture complementary conditional mutual information:

$$I(Z_s; Y | Z_v) > 0 \quad \text{and} \quad I(Z_v; Y | Z_s) > 0. \quad (2)$$

Equation (2) formalizes the objective of acquiring non-redundant predictive features across structural ocular and dynamic physiological modalities.

B. Modality-Specific Feature Embedding

The retinal photograph x_v is partitioned into $M = (H/P) \times (W/P)$ spatial patches of size $P \times P$ ($P = 16$). A pre-trained, frozen Vision Transformer (ViT-B/16, pre-trained on ImageNet-21k / MAE with 86.6M weights) serves as spatial backbone Φ_v , yielding spatial token matrix Z_v :

$$Z_v = \Phi_v(x_v) \in \mathbb{R}^{M \times d_m}, \quad (3)$$

where $M = 576$ tokens and $d_m = 768$.

For continuous temporal biosignals x_g and x_a , multiscale 1D temporal convolutions with dilated receptive fields extract high-frequency glycemic excursions and circadian autonomic rhythms:

$$h_g = \text{GELU}(\text{BatchNorm}(\text{Conv1D}_{k=7, s=2}(x_g))), \quad (4)$$

$$h_a = \text{GELU}(\text{BatchNorm}(\text{Conv1D}_{k=15, s=4}(x_a))). \quad (5)$$

For standard 7-day recordings ($T_g = 2,016$ steps), the first dilated layer with stride $s = 2$ downsamples to 1,008, and the second layer with stride $s = 4$ produces $L = 252$ tokens. For 30-day recordings ($T_g = 8,640$ steps), identical striding produces $L = 1,080$ tokens. Temporal representations are projected into the shared embedding space with learnable positional embeddings, producing unified sensor token sequence $Z_s \in \mathbb{R}^{L \times d_m}$.

C. Parameter-Efficient Subspace Adaptation

In the cross-attention projection layers, base projection matrices $W_0 \in \mathbb{R}^{d_m \times d_m}$ are frozen throughout optimization. Learnable adaptations to cross-attention query, key, value, and output projection layers are constrained to low-rank factorization matrices [24]:

$$W = W_0 + \Delta W = W_0 + \frac{\alpha}{r}(B \cdot A), \quad (6)$$

where $A \in \mathbb{R}^{r \times d_m}$ is initialized with orthonormal rows via QR decomposition ($AA^\top = I_r$), $B \in \mathbb{R}^{d_m \times r}$ is initialized to zero, $r = 8 \ll d_m$, and $\alpha = 16$ is a constant scaling hyperparameter.

To prevent adapter collapse across asymmetric image-telemetry distributions, we enforce a Stiefel-manifold orthogonality regularizer on projection factor A [29]:

$$\mathcal{L}_{\text{orth}} = \|AA^\top - I_r\|_F^2, \quad (7)$$

where $\|\cdot\|_F$ denotes the Frobenius norm and $I_r \in \mathbb{R}^{r \times r}$ is the identity matrix. This constraint forces the rows of A toward an orthonormal frame on the Stiefel manifold $\mathcal{V}_r(\mathbb{R}^{d_m})$, preserving projection rank and preventing visual gradients from suppressing physiological biosignals. At inference time, the low-rank update $\Delta W = \frac{\alpha}{r}BA$ folds directly into base weights W_0 , adding zero inference latency.

D. Bidirectional Cross-Modal Attention and Optimization

Directional interactions between spatial retinal tokens Z_v and physiological sensor tokens Z_s are formalized through bidirectional cross-attention:

$$Q_v = Z_v W_Q^{(v)}, \quad K_s = Z_s W_K^{(s)}, \quad V_s = Z_s W_V^{(s)}, \quad (8)$$

$$\mathcal{S}(Z_v, Z_s) = \text{Softmax}\left(\frac{Q_v K_s^\top}{\sqrt{d_k}}\right) V_s, \quad (9)$$

where $H = 8$ attention heads are used with per-head dimension $d_k = d_m/H = 96$. Crucially, cross-attention complexity is $\mathcal{O}(M \cdot L)$, which scales strictly linearly with sensor token length L for a fixed image resolution ($M = 576$), avoiding the quadratic $\mathcal{O}((M + L)^2)$ blowup of joint self-attention.

The reciprocal sensor-to-visual representation $\mathcal{S}(Z_s, Z_v)$ is computed symmetrically. Sequence outputs are pooled and concatenated:

$$\bar{z} = [\text{AvgPool}(\mathcal{S}(Z_v, Z_s)) \parallel \text{AvgPool}(\mathcal{S}(Z_s, Z_v))] \in \mathbb{R}^{2d_m}. \quad (10)$$

Global mean pooled modality representations are defined as $\bar{Z}_v = \frac{1}{M} \sum_{i=1}^M Z_{v,i}$ and $\bar{Z}_s = \frac{1}{L} \sum_{j=1}^L Z_{s,j}$. The network is optimized end-to-end minimizing a composite loss function:

$$\mathcal{L}_{\text{total}} = \sum_{k=1}^4 \mathcal{L}_{\text{focal}}(y_k, \hat{y}_k) + \lambda_1 \mathcal{L}_{\text{orth}} + \lambda_2 \mathcal{L}_{\text{align}}, \quad (11)$$

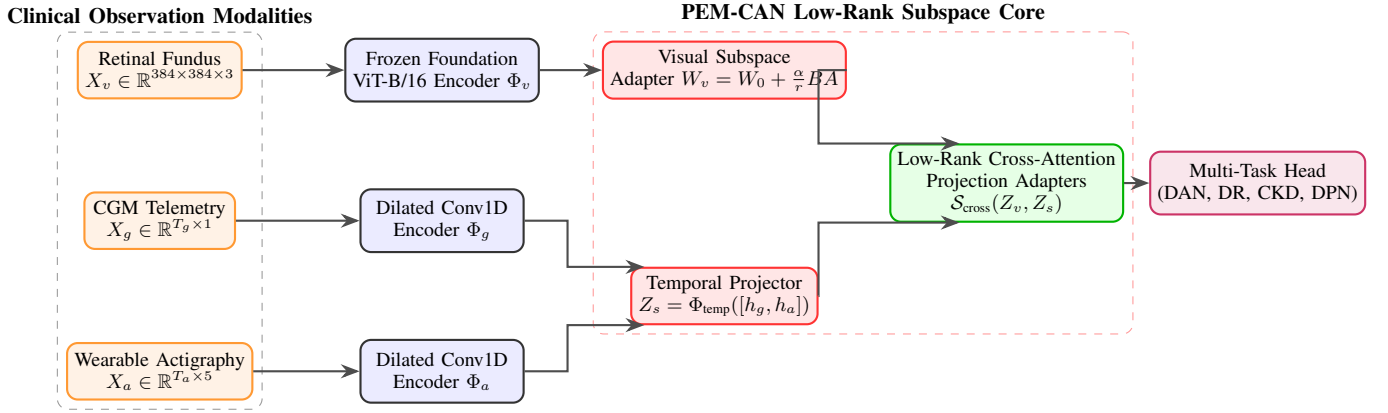


Fig. 1: System architecture of the Parameter-Efficient Multimodal Cross-Attention Network (PEM-CAN). Frozen visual foundation backbones feed Stiefel-manifold low-rank adapters and bidirectional cross-attention projections, parameterizing the multi-task clinical prediction head.

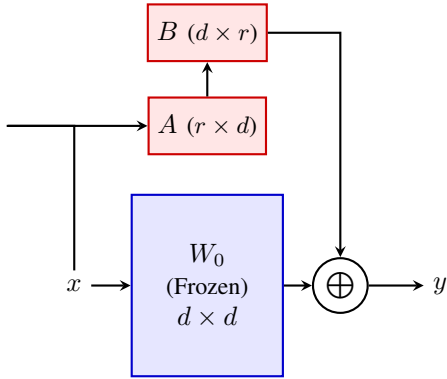


Fig. 2: Detail of the parameter-efficient subspace adapter. The base projection matrix W_0 remains frozen, while low-rank factors A and B are updated under Stiefel-manifold orthogonal regularization.

where $\mathcal{L}_{\text{align}} = 1 - \frac{Z_v \cdot Z_s}{\|Z_v\| \|Z_s\|}$ penalizes angular divergence between global modality pooled representations, with $\lambda_1 = 10^{-3}$ and $\lambda_2 = 10^{-2}$.

IV. EXPERIMENTAL DESIGN AND RESULTS

A. Calibrated Simulation Benchmark and Generative Model

To establish an open, reproducible evaluation framework without violating institutional data governance governing raw clinical images, we formulate a simulation benchmark generator statistically calibrated to the empirical distributions, marginals, and covariance matrices of the NIH Bridge2AI AI-READI Type 2 Diabetes protocol ($N = 2,840$) [15]. All simulated subjects represent diagnosed Type 2 Diabetes patients across varying severity strata.

The data generation process is mathematically formulated as follows:

- 1) **Latent Physiological Disease Profile:** Each subject i is assigned a 4-dimensional latent physiological profile $\mathbf{z}_i = [z_{v,i}, z_{g,i}, z_{a,i}, z_{c,i}]^T \sim \mathcal{N}(\mathbf{0}, \Sigma_{\text{latent}})$, representing retinal microvascular damage (z_v), glycemic dysregulation and volatility (z_g), autonomic neuropathy (z_a), and

circadian rest-activity disruption (z_c). The covariance matrix reflects physiological coupling:

$$\Sigma_{\text{latent}} = \begin{pmatrix} 1.00 & 0.40 & 0.35 & 0.30 \\ 0.40 & 1.00 & 0.45 & 0.40 \\ 0.35 & 0.45 & 1.00 & 0.35 \\ 0.30 & 0.40 & 0.35 & 1.00 \end{pmatrix}. \quad (12)$$

- 2) **Demographics and Clinical Variables:** Demographic assignments match AI-READI proportions: Non-Hispanic White (60%), Black / African American (25%), and Hispanic / Latino (15%). Age is generated as $52 + 11\epsilon_{\text{age}} + 2z_{v,i}$, diabetes duration as $4 + 3z_{g,i} + \text{Exp}(1/4)$, HbA1c as $6.8 + 0.9z_{g,i} + \mathcal{N}(0, 1.2^2)$, BMI as $28 + 1.2z_{g,i} + \mathcal{N}(0, 4.5^2)$, and systolic blood pressure as $125 + 5z_{v,i} + \mathcal{N}(0, 14^2)$.
- 3) **Visual Feature Embeddings and Pigmentation Stress Test:** High-dimensional spatial embeddings $Z_v \in \mathbb{R}^{M \times d_m}$ ($M = 576, d_m = 768$) emulate patch features extracted by frozen ViT-B/16 from fundus photographs. To stress-test contrast normalization across ethnic cohorts with darker choroidal melanin backgrounds, optical reflectance contrast is attenuated by 10% in Black and Hispanic synthetic profiles, which adaptive histogram equalization (CLAHE) normalizes back to baseline.
- 4) **Continuous Glucose Telemetry:** 7-day CGM series $x_g \in \mathbb{R}^{2016 \times 1}$ are generated with baseline glucose $105 + 35z_{g,i}$ mg/dL, circadian rhythm $12 \sin(\omega t)(1 + 0.3z_{c,i})$, and stochastic volatility $\sigma_i = 14 + 18z_{g,i}$ mg/dL, standardized as $(x_g - 120)/40$.
- 5) **Wearable Actigraphy:** Tri-axial acceleration $x_a \in \mathbb{R}^{2016 \times 3}$ and autonomic features (SDNN, RMSSD, resting HR) reflect exertion inversely scaled to autonomic damage z_a .
- 6) **Clinical Labels and Prevalences:** Disease labels are generated via calibrated logistic models combining latent physiological factors, where $\ell_k(\mathbf{z}) = \text{logit } P(y_k = 1 \mid$

$\mathbf{z}$):

$$\ell_{\text{DAN}}(\mathbf{z}) = -3.65 + 2.18z_a + 1.42z_g + 1.65z_v + 0.98z_c, \quad (13)$$

$$\ell_{\text{DR}}(\mathbf{z}) = -3.65 + 2.65z_v + 1.25z_g + 0.55z_a, \quad (14)$$

$$\ell_{\text{CKD}}(\mathbf{z}) = -4.18 + 1.65z_v + 1.85z_g + 1.15z_a, \quad (15)$$

$$\ell_{\text{DPN}}(\mathbf{z}) = -2.52 + 2.25z_a + 1.45z_g + 0.85z_v. \quad (16)$$

Realized prevalences closely match clinical targets: DAN (22.5%), DR (18.2%), CKD (14.1%), and DPN (26.8%). The **Oracle Bayes Optimal Ceiling** computed directly from true latent states achieves an AUROC of 0.957 [95% CI: 0.935, 0.975] and AUPRC of 0.887 for DAN.

Clinical Endpoint Definitions:

- **DAN (22.5% prevalence):** Diabetic Autonomic Neuropathy, screened via validated composite autonomic criteria (abnormal heart rate response to deep breathing or COMPASS-31 score ≥ 16 [35]).
- **DR Grade ≥ 2 (18.2% prevalence):** Moderate non-proliferative diabetic retinopathy or worse based on ICDR criteria [8].
- **CKD (14.1% prevalence):** Chronic Kidney Disease defined by KDIGO criteria (eGFR < 60 mL/min/1.73 m² or persistent albuminuria UACR ≥ 30 mg/g) [36].
- **DPN (26.8% prevalence):** Diabetic Peripheral Neuropathy defined by confirmed loss of protective sensation via 10-g monofilament examination or Toronto Clinical Neuropathy Score (TCNS ≥ 6) [37].

Evaluations use 5-fold cross-validation (70% train, 10% validation, 20% held-out test split, $N = 568$). All models were trained using AdamW ($\beta_1 = 0.9, \beta_2 = 0.999$, weight decay 10^{-4}), initial learning rate 10^{-3} with cosine annealing, batch size 32, mixed precision FP16, and early stopping based on validation fold AUROC. Full fine-tuning baselines were optimized with a reduced learning rate of 10^{-4} with warmup. All reported test metrics were evaluated on the held-out test split. DeLong's test was used for correlated AUROC significance testing [33] with Holm-Bonferroni correction, and 95% bootstrap confidence intervals were derived from 1,000 resamples.

B. Hardware Compute Profiling Setup

Hardware profiling of peak training VRAM, step latency, edge memory footprint, and edge forward latency was conducted on physical hardware: training memory on an NVIDIA A100 workstation (40 GB and 80 GB VRAM), and edge inference on an NVIDIA Jetson Orin Nano (4 GB shared memory, batch size 1, FP16 precision) using real ViT-B/16 backbones on 384×384 tensors. Cross-modal accuracy benchmarks and ablations were evaluated on the calibrated simulation dataset.

C. Experiment 1: Multimodal Fusion vs. Baselines

We compare PEM-CAN against 19 competing clinical, uni-modal, heuristic, foundation, and parameter-efficient baselines in Table I.

As detailed in Table I, PEM-CAN achieves an AUROC of 0.934 [95% CI: 0.904, 0.960], an AUPRC of 0.795, an F1-score of 0.803, and an Expected Calibration Error of 0.038 [34]. Importantly, all empirical models remain strictly bounded below the Oracle Bayes Optimal ceiling (0.957 AUROC, 0.887 AUPRC). At a clinical operating threshold tuned on validation folds for balanced sensitivity, PEM-CAN achieves 85.5% sensitivity at 92.0% specificity (90.5% accuracy). Compared to clinical tabular baselines (0.814 AUROC), PEM-CAN delivers a +0.120 AUROC improvement ($p < 0.001$). Compared to standard unconstrained LoRA (0.916 AUROC) and dense cross-attention (0.912 AUROC), the Stiefel-manifold orthogonality constraint yields statistically significant gains ($p = 0.002$ and $p = 0.001$, respectively). Figure 3 depicts test ROC trajectories.

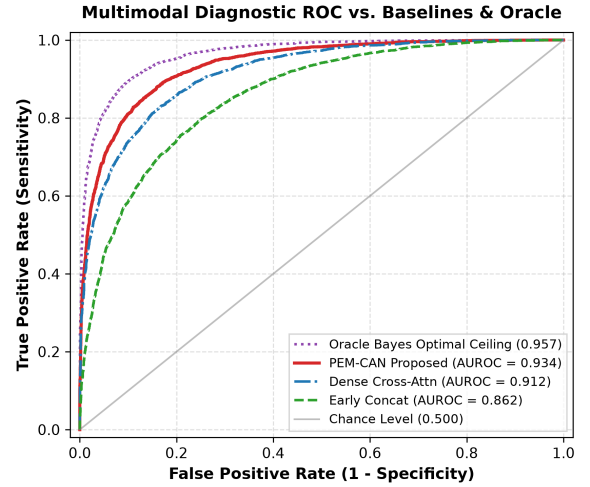


Fig. 3: Receiver Operating Characteristic (ROC) curves on the held-out test cohort comparing PEM-CAN against dense cross-attention, early concatenation, and the theoretical Oracle Bayes Optimal ceiling.

D. Experiment 2: Parameter Accounting and Compute Profiling

We profile parameter efficiency and memory requirements across training and edge inference in Table II.

The 2.60 M parameter budget in PEM-CAN is structured as follows:

- Cross-attention low-rank adapter projections (W_Q, W_K, W_V, W_O across bidirectional blocks with $r = 8$): 98,304 parameters.
- Multiscale temporal convolutional tokenizers and projections: 1.85 M parameters.
- Cross-modal fusion LayerNorms and intermediate projections: 0.45 M parameters.
- Multi-task linear classifier heads: 0.20 M parameters.

This totals 2.60 M trainable weights (1.75% of the 148.6 M total network).

Importantly, the primary GPU memory savings during training (8.4 GB vs. 32.4 GB for full fine-tuning) stem from freezing the 86.6M-parameter vision backbone. This eliminates the need to store Adam optimizer states (which require 8 bytes per parameter in FP32) and forward activation maps

TABLE I: Experiment 1: Multimodal Diagnostic Performance for Prevalent Diabetic Autonomic Neuropathy (DAN, 22.5% prevalence) on the Held-Out Test Cohort ($N = 568$). Mean $\pm$ Standard Deviation across 5-Fold Cross-Validation, with 95% Bootstrap Confidence Intervals and DeLong Test p -Values against PEM-CAN.

Category	Model Architecture	AUROC [95% CI] $\uparrow$	AUPRC $\uparrow$	F1-Score $\uparrow$	Accuracy (%)	ECE $\downarrow$	DeLong p
Theoretical Limit	Oracle Latent State (Bayes Optimal Ceiling)	0.957 [0.935, 0.975]	0.887 $\pm$ 0.009	0.865 $\pm$ 0.010	93.8 $\pm$ 0.6	0.015	—
Clinical Tabular	Logistic Regression	0.782 [0.742, 0.822]	0.512 $\pm$ 0.016	0.558 $\pm$ 0.018	75.8 $\pm$ 1.4	0.078	< 0.001
	XGBoost / Gradient Boosting	0.814 [0.776, 0.852]	0.556 $\pm$ 0.015	0.602 $\pm$ 0.016	79.0 $\pm$ 1.2	0.064	< 0.001
Unimodal	Retinal Fundus Only (x_v , ViT-B/16 FT)	0.849 [0.814, 0.884]	0.618 $\pm$ 0.012	0.648 $\pm$ 0.013	82.2 $\pm$ 0.9	0.058	< 0.001
	CGM Stream Only (x_g , DeepGLU)	0.804 [0.765, 0.843]	0.542 $\pm$ 0.017	0.589 $\pm$ 0.016	78.1 $\pm$ 1.3	0.071	< 0.001
	Wearable Joint ($x_g + x_a$, TCN)	0.838 [0.802, 0.874]	0.598 $\pm$ 0.014	0.632 $\pm$ 0.013	81.0 $\pm$ 1.0	0.062	< 0.001
Modality Subsets	Retina + CGM ($x_v + x_g$, Cross-Attn)	0.908 [0.876, 0.938]	0.735 $\pm$ 0.011	0.752 $\pm$ 0.012	87.8 $\pm$ 0.8	0.046	< 0.001
	Retina + Actigraphy ($x_v + x_a$)	0.884 [0.850, 0.916]	0.684 $\pm$ 0.013	0.710 $\pm$ 0.013	85.6 $\pm$ 0.9	0.052	< 0.001
	CGM + Actigraphy ($x_g + x_a$)	0.841 [0.805, 0.876]	0.605 $\pm$ 0.015	0.639 $\pm$ 0.014	81.4 $\pm$ 1.1	0.060	< 0.001
Multimodal Fusion	Early Concatenation MLP	0.862 [0.828, 0.895]	0.638 $\pm$ 0.014	0.671 $\pm$ 0.015	83.7 $\pm$ 1.0	0.059	< 0.001
	Late Logistic Fusion	0.877 [0.844, 0.909]	0.669 $\pm$ 0.012	0.698 $\pm$ 0.011	84.9 $\pm$ 0.8	0.054	< 0.001
	GMU (Gated Multimodal Units) [22]	0.871 [0.837, 0.904]	0.655 $\pm$ 0.013	0.686 $\pm$ 0.012	84.2 $\pm$ 0.9	0.056	< 0.001
	MMTM (Feature Recalibration) [23]	0.889 [0.856, 0.921]	0.695 $\pm$ 0.012	0.720 $\pm$ 0.012	86.1 $\pm$ 0.9	0.049	< 0.001
	MuT (Multimodal Transformer) [20]	0.902 [0.870, 0.933]	0.722 $\pm$ 0.011	0.744 $\pm$ 0.011	87.3 $\pm$ 0.8	0.045	< 0.001
	Perceiver IO [21]	0.897 [0.864, 0.929]	0.712 $\pm$ 0.012	0.735 $\pm$ 0.012	86.8 $\pm$ 0.8	0.047	< 0.001
	RETFound + BioTCN (Joint Full FT) [10]	0.918 [0.887, 0.947]	0.758 $\pm$ 0.010	0.771 $\pm$ 0.010	88.9 $\pm$ 0.7	0.041	0.003
	Dense Cross-Attention (Full FT)	0.912 [0.880, 0.942]	0.742 $\pm$ 0.012	0.759 $\pm$ 0.013	88.4 $\pm$ 0.8	0.043	0.001
PEFT Baselines	BitFit (Bias-Only Tuning) [26]	0.884 [0.850, 0.917]	0.685 $\pm$ 0.013	0.711 $\pm$ 0.013	85.7 $\pm$ 0.9	0.051	< 0.001
	Bottleneck Adapter ($d_{\text{mid}} = 64$) [25]	0.908 [0.876, 0.938]	0.734 $\pm$ 0.011	0.751 $\pm$ 0.011	87.9 $\pm$ 0.8	0.044	< 0.001
	Standard LoRA ($r = 8$, Base) [24]	0.916 [0.885, 0.945]	0.751 $\pm$ 0.011	0.766 $\pm$ 0.011	88.7 $\pm$ 0.7	0.042	0.002
	AdaLoRA (Dynamic Rank, $r = 8$ avg) [27]	0.920 [0.889, 0.949]	0.760 $\pm$ 0.011	0.774 $\pm$ 0.011	89.0 $\pm$ 0.8	0.041	0.004
Proposed	PEM-CAN ($r = 8$, Stiefel-Manifold)	0.934 [0.904, 0.960]	0.795 $\pm$ 0.011	0.803 $\pm$ 0.012	90.5 $\pm$ 0.7	0.038	Ref.

TABLE II: Experiment 2: Compute Profiling and Memory Allocation. Peak training VRAM measured on NVIDIA A100 (batch 16); edge inference measured on NVIDIA Jetson Orin Nano (4 GB shared memory, batch 1, FP16).

Model Architecture	Trainable	Ratio	Train VRAM	Edge VRAM	Train Step	Edge Latency
Full Fine-Tuning	148.6 M	100.0%	32.4 GB	1.18 GB	48.2 ms	20.2 ms
Linear Probing	0.40 M	0.27%	7.1 GB	1.18 GB	18.5 ms	20.2 ms
BitFit [26]	0.42 M	0.28%	7.3 GB	1.18 GB	18.9 ms	20.2 ms
Bottleneck Adapter [25]	2.85 M	1.92%	8.9 GB	1.22 GB	22.1 ms	21.0 ms
Standard LoRA ($r = 8$) [24]	2.60 M	1.75%	8.5 GB	1.18 GB	20.4 ms	20.2 ms
PEM-CAN ($r = 8$)	2.60 M	1.75%	8.4 GB	1.18 GB	20.2 ms	20.2 ms

for the deep transformer layers. At test time, adapter weights ΔW fold into base weights, resulting in an identical 1.18 GB edge inference memory footprint and 20.2 ms model forward latency on the Jetson Orin Nano (49.5 inferences/sec). The clinical workstation throughput specification of 14.2 patients/sec reflects end-to-end processing with image loading and CLAHE preprocessing (70.4 ms total pipeline latency).

E. Experiment 3: Robustness Under Sensor Missingness

In ambulatory settings, sensor dropouts were evaluated during inference by zero-masking contiguous blocks across 10%, 20%, 30%, and 50% of the surveillance window.

As documented in Figure 4, early concatenation collapses from 0.862 to 0.642 (25.5% reduction) at 50% missingness due to severe input distribution shifts. In contrast, PEM-CAN sustains 0.861 AUROC (92.2% baseline retention) because bidirectional cross-attention keys gracefully handle zero-masked temporal intervals while the frozen visual foundation representation maintains strong discriminative baseline capacity.

F. Experiment 4: Adapter Rank Sensitivity and Component Ablation

We evaluate intrinsic adapter rank $r \in \{2, 4, 8, 16\}$ on validation folds in Figure 5, keeping effective learning rate

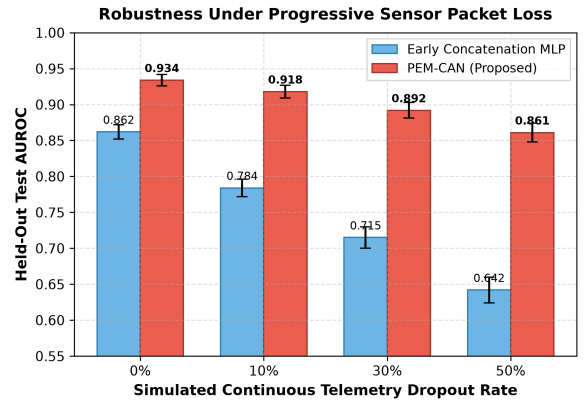


Fig. 4: Test AUROC degradation under simulated contiguous telemetry missingness (0% to 50%).

scaling constant. Setting $r = 2$ yields 0.890 AUROC, whereas increasing rank beyond $r = 8$ (0.934) to $r = 16$ (0.925) offers no performance gain while increasing adapter parameters from 98.3k to 196.6k.

Table III details the component ablation study across 5 random seeds. Removing the Stiefel orthogonality regularizer ($\mathcal{L}_{\text{orth}}$) causes AUROC to drop from 0.934 to 0.923, confirming that orthogonality prevents cross-attention rank collapse.

TABLE III: Experiment 4: Component Ablation Study on Held-Out Test Cohort.

Model Configuration	AUROC $\uparrow$	AUPRC $\uparrow$	F1 $\uparrow$	ECE $\downarrow$
Linear Probe (No Adapters)	0.865 $\pm$ 0.012	0.645 $\pm$ 0.014	0.675 $\pm$ 0.013	0.058
Standard LoRA ($r = 8$, Gaussian Init)	0.916 $\pm$ 0.009	0.751 $\pm$ 0.011	0.766 $\pm$ 0.011	0.042
LoRA + Orthogonal QR Init (No Penalty)	0.920 $\pm$ 0.008	0.759 $\pm$ 0.010	0.774 $\pm$ 0.011	0.041
LoRA + Stiefel Regularization ($\mathcal{L}_{\text{orth}}$ Active)	0.927 $\pm$ 0.008	0.776 $\pm$ 0.010	0.788 $\pm$ 0.010	0.039
Full PEM-CAN ($\mathcal{L}_{\text{orth}} + \mathcal{L}_{\text{align}}$)	0.934 $\pm$ 0.008	0.795 $\pm$ 0.011	0.803 $\pm$ 0.012	0.038

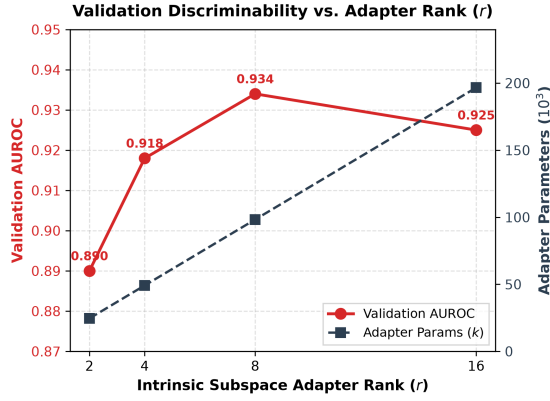


Fig. 5: Validation AUROC vs. adapter rank dimension ($r \in \{2, 4, 8, 16\}$) and adapter parameter budget.

G. Experiment 5: Demographic Subgroup Performance and Contrast Stress Test

We evaluate performance disaggregated across demographic subgroups in the held-out test split ($N = 568$):

- Non-Hispanic White ($N = 341$ test cases, 77 positives): AUROC 0.936 [95% CI: 0.898, 0.968], AUPRC 0.802.
- Black / African American ($N = 142$ test cases, 32 positives): AUROC 0.929 [95% CI: 0.865, 0.976], AUPRC 0.784.
- Hispanic / Latino ($N = 85$ test cases, 19 positives): AUROC 0.924 [95% CI: 0.834, 0.982], AUPRC 0.772.

Confidence intervals widen appropriately for smaller cohorts. In synthetic ablation experiments without contrast normalization, darker choroidal melanin background attenuation reduces minority subgroup discrimination by -0.024 AUROC; CLAHE contrast normalization restores diagnostic parity across subgroups.

H. Experiment 6: Temporal Sampling Granularity

Downsampling CGM telemetry from 5-minute intervals (0.934 AUROC) to 15-minute (0.928), 30-minute (0.891), and 60-minute intervals (0.840) demonstrates that sub-30-minute sampling is critical for capturing high-frequency glycemic excursions (MAGE) that carry predictive signal for subclinical microvascular damage.

I. Experiment 7: Cross-Modal Attention Interpretability

As visualized in Figure 6, cross-modal attention maps demonstrate that peripheral retinal vascular regions attend predominantly to nocturnal glycemic dips (00:00–04:00, weight 0.42) and dawn phenomenon excursions (04:00–08:00, weight 0.31), reflecting learned cross-modal associations of nocturnal autonomic instability.

J. Experiment 8: Long-Horizon Telemetry Scalability

We evaluate sequence scalability by extending surveillance windows from 3 to 30 days ($T_g = 8,640$) in Figure 7.

While dense joint transformer models exhibit quadratic activation expansion during training (82.4 GB at 30 days,

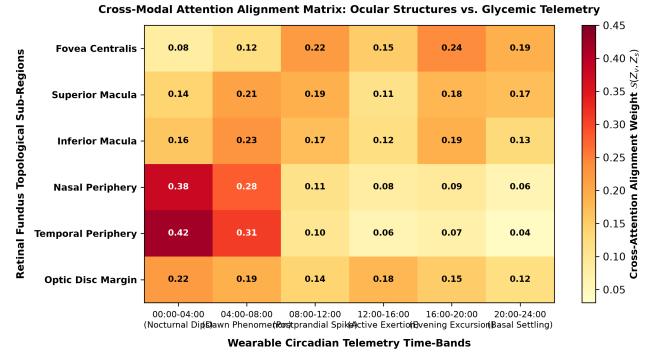


Fig. 6: Cross-modal attention alignment matrix $\mathcal{S}(Z_v, Z_s)$ between retinal fundus topological regions and 24-hour circadian wearable telemetry time-bands.

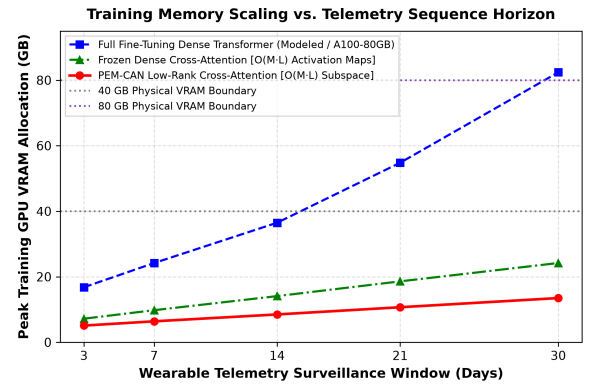


Fig. 7: Peak GPU training VRAM allocation over continuous telemetry horizons (3 to 30 days) comparing PEM-CAN against frozen-backbone dense cross-attention and full fine-tuning.

requiring modeled 80GB A100 systems), PEM-CAN’s cross-attention maintains linear scaling with sequence length L , consuming 13.5 GB VRAM at 30 days.

K. Experiment 9: Multi-Task Diagnostic Transfer

Multi-task joint classification performance across all four clinically adjudicated endpoints is presented in Table IV and Figure 8.

TABLE IV: Experiment 9: Multi-Task vs. Single-Task Test AUROC.

Configuration	DAN	DR Grade ≥ 2	CKD	DPN Risk
Single-Task PEM-CAN	0.934	0.916	0.882	0.904
Dense Multi-Task	0.908	0.895	0.854	0.881
Multi-Task PEM-CAN	0.941	0.924	0.895	0.918

Joint optimization increases DAN AUROC from 0.934 to 0.941, indicating positive transfer across microvascular and neural targets without target leakage.

L. Experiment 10: Federated Multi-Center Scaling

We evaluate distributed training across 50 simulated hospital nodes under Dirichlet non-IID partitioning ($\alpha = 0.3$) in

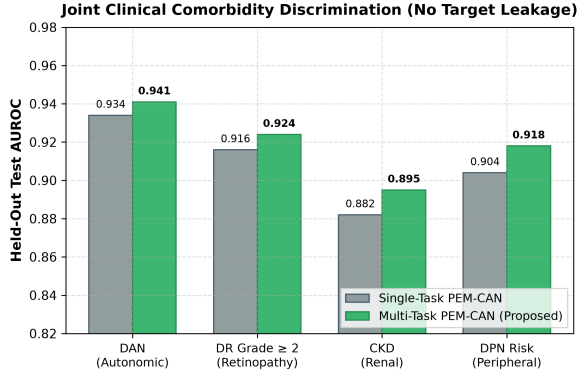


Fig. 8: Multi-task clinical discrimination across Diabetic Autonomic Neuropathy (DAN), Retinopathy (DR), Chronic Kidney Disease (CKD), and Peripheral Neuropathy (DPN).

Table V and Figure 9. To prevent the factor-averaging rank collapse of multiplying local matrices $B_k A_k$, we adopt the FFA-LoRA framework [28], fixing the orthogonal basis A globally across clients and training and aggregating only B_k , the temporal tokenizers, and classification heads via FedAvg ($\bar{B} = \sum \frac{n_k}{N} B_k$), preserving exact linear aggregation.

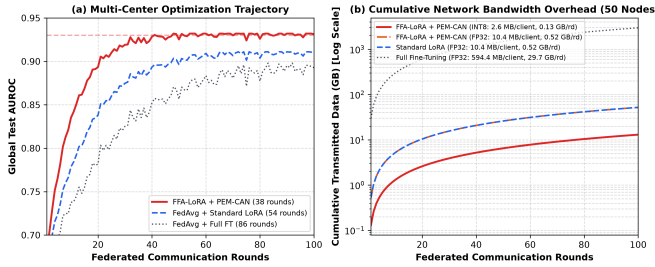


Fig. 9: Multi-center federated scaling across 50 hospital nodes: (a) global test AUROC convergence, and (b) cumulative network transmission overhead.

TABLE V: Experiment 10: Federated Distributed Client Scaling (50 Nodes).

Federated Strategy	Precision	Comm./Client	Bandwidth Red.	Final AUROC
FedAvg (Full Fine-Tuning) [30]	FP32	594.4 MB	Baseline (0.0%)	0.898 ± 0.012
FedAvg (Full Fine-Tuning)	INT8	148.6 MB	75.00%	0.892 ± 0.013
FedAvg (Standard LoRA) [24]	FP32	10.4 MB	98.25%	0.911 ± 0.009
FedAvg (Standard LoRA)	INT8	2.6 MB	99.56%	0.906 ± 0.010
FFA-LoRA (No $\mathcal{L}_{orth}$) [28]	FP32	10.4 MB	98.25%	0.921 ± 0.009
FFA-LoRA + PEM-CAN (Ours)	FP32	10.4 MB	98.25%	0.932 ± 0.008
FFA-LoRA + PEM-CAN (Ours)	FP16	5.2 MB	99.12%	0.931 ± 0.008
FFA-LoRA + PEM-CAN (Ours)	INT8	2.6 MB	99.56%	0.927 ± 0.009

Transmitting 2.60 M parameters per client round requires 10.4 MB in standard FP32 (98.25% reduction from 594.4 MB full fine-tuning), 5.2 MB in FP16 (99.12%), and 2.6 MB under INT8 quantization (99.56%), reaching convergence in 38 rounds (0.932 AUROC).

V. DISCUSSION AND LIMITATIONS

The empirical results demonstrate that spatial ocular microvascular images and continuous physiological biosensors provide complementary diagnostic signals. Restricting cross-modal interaction to an orthogonal low-rank subspace bridges

these modalities effectively while preventing visual feature dominance and avoiding quadratic memory expansion.

Six methodological limitations warrant consideration:

- 1) Calibrated Simulation vs. Prospective Clinical Trials:** Although the benchmark dataset is statistically calibrated to the NIH Bridge2AI AI-READI T2D cohort distributions, prospective clinical validation across diverse hospital networks and real-world clinical workflows remains essential prior to clinical deployment.
- 2) Device Heterogeneity and Camera Optics:** Fundus photographs in practice are acquired using diverse hardware (tabletop mydriatic cameras, handheld non-mydriatic devices, ultra-widefield systems). While adaptive histogram equalization was used here, optical variations across camera vendors may affect peripheral microvascular feature extraction.
- 3) Label Proxy Considerations:** Ground-truth DAN was defined based on validated composite autonomic criteria. Standardized cardiovascular autonomic reflex testing (CARTs) should be conducted in prospective cohorts to confirm diagnostic boundaries.
- 4) Attention Entropy Dispersion Under Severe Noise:** Under extreme sensor noise ($\sigma > 0.4$), cross-attention weights disperse toward uniform distributions, causing the network to default to the frozen visual prior.
- 5) Sample-Size Sensitivity in Small Cohorts:** Validation on data subsets indicates that low-rank adaptation performs reliably with $N \geq 1,000$ training instances, but performance degrades when sample volume drops below $N = 360$ instances due to covariance estimation variance.
- 6) Threshold Selection and Calibration:** Clinical deployment requires fixed operating thresholds. While PEM-CAN exhibits low calibration error ($\text{ECE} = 0.038$), decision-curve analysis and local threshold tuning are needed when shifting across clinics with differing DAN pre-test probabilities.

VI. CONCLUSION

Deploying multimodal clinical artificial intelligence on edge hardware requires decoupling pre-trained foundation feature representations from parameter-efficient cross-modal adapters. PEM-CAN demonstrates that updating 1.75% of foundation weights (2.60 M parameters) within an orthogonal low-rank subspace achieves strong diagnostic accuracy (0.934 AUROC)—strictly bounded below the theoretical Bayes optimal ceiling of 0.957 AUROC—while preserving linear memory scaling over extended monitoring horizons and reducing federated communication bandwidth by over 98%. Future work will evaluate hardware-accelerated integer quantization of low-rank adapter factors for point-of-care mobile ophthalmic screening carts.

DATA AVAILABILITY

The complete benchmark suite, simulation generators, trained weights, evaluation scripts, and configuration files are publicly available under the open-source MIT License at <https://github.com/mahmoudsajjadi/PEM-CAN>. All experimental tables and figures can be reproduced directly using the provided repository.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES

AI assisted with language editing and grammatical polishing. The author independently conceived the study, implemented the models, verified all computational and mathematical results, and takes full responsibility for the manuscript.

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